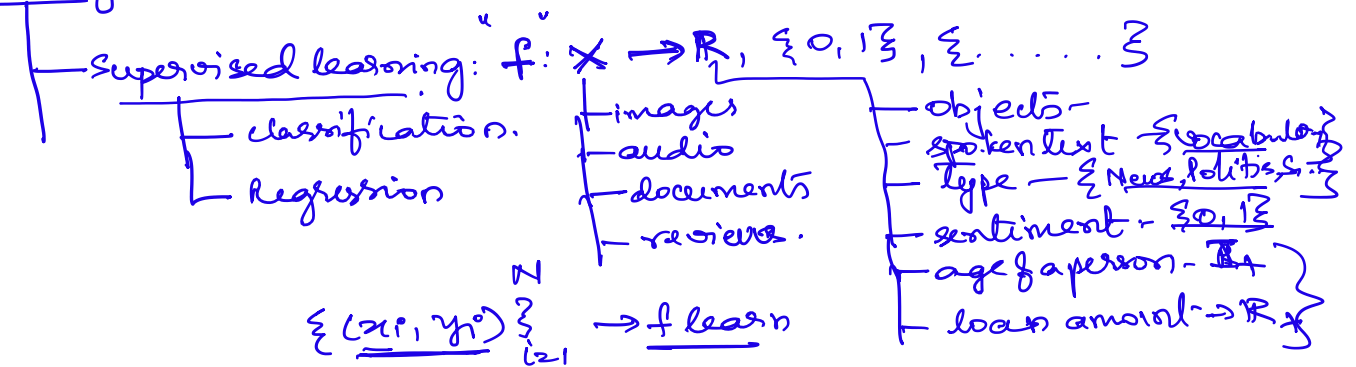


learning:



representation

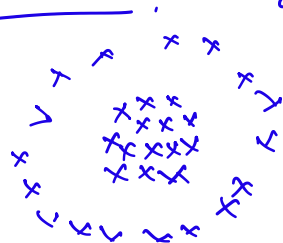
- f - line - slope, intercept.
- circle - center, radius
- rectangle - top left, bottom right.



Evaluate the goodness. $L(f(x_i), y_i)$ - loss function

Optimization: find the parameters θ which minimize the loss function L , given $\{ (x_i, y_i) \}_{i=1}^N$.

Unsupervised learning:

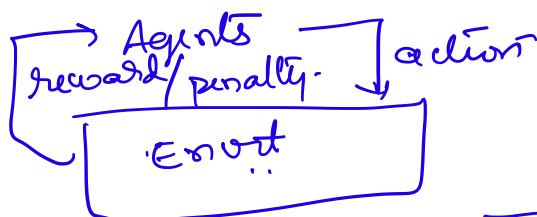


$\{ x_i \}_{i=1}^N$

clusters / groups

$f: X \rightarrow \text{unknown}$

Reinforcement learning:



if $c_1, \boxed{c_2}, c_3, c_4$ then O_1

if c_1, c_3, c_4 then O_1